# A Family and selection-rule comparisons

All panels use the same 305 recordings and two-rater references. Model statistics average agreement with the three possible two-human means; human consensus compares each human with the other two. Marginal intervals retain the 2,000-draw, seed-42 evaluation export. S and D use the 1-grid $\{0.001, 0.01, 0.1, 1, 10, 100\}$, selected respectively by construct-specific Spearman and accuracy MAE. H uses accuracy MAE on the 5-grid $\{0.00005, 0.0005, 0.005, 0.05, 0.5, 5\}$. The 5-grid converts the earlier summed-loss candidates $\{0.1, 1, 10, 100, 1000, 10000\}$ by dividing by the original 2,000 SpeechOcean training utterances; the same fixed grid is used for every family. The conversion preserves penalty scale, rather than reproducing every earlier implementation detail. All fitting, scaling, and tuning use public data only.

Three-ratio banded ridge searches XLS-R/XEUS penalty ratios $r \in \{1/4, 1, 4\}$ for fluency and prosody. Its expanded scalar grid combines $10^{-6}, 10^{-5.5}, \ldots, 10^{3}$ with every 1-grid and 5-grid value. For each rule and construct, the equal-penalty winner is additionally refined over nine logarithmically spaced points from $\alpha/\sqrt{10}$ to $\alpha\sqrt{10}$, together with the expanded candidates. The banded search includes that refined equal-penalty solution as a fallback. Relative feature gains $r^{1/4}$ for XEUS and $r^{-1/4}$ for XLS-R yield penalties $\alpha/\sqrt{r}$ and $\alpha\sqrt{r}$ in the original standardized coordinates. B-S selects construct Spearman, breaking ties by lower MAE, equal penalties, then larger scalar penalty; B-D selects accuracy MAE, breaking ties by equal penalties, then larger scalar penalty. Accuracy has no banded search and retains its corresponding refined scalar-ridge head. The broadened scalar grid and unequal-penalty search change together, so the comparison does not isolate banding alone. The external fluency leader B-S-aesopocean uses scalar $\alpha = 10$ and ratio $r = 4$ (XEUS/XLS-R penalties 5 and 20); the prosody-correlation leader B-D-koreanaesop uses $\alpha = 5$, $r = 4$ (penalties 2.5 and 10). These settings were selected on public development data.

All 35 configurations completed all 305 recordings. The original S/D/H arrays are unchanged; the 14 B variants add the exact evaluated checkpoints. No private ratings entered fitting, tuning, or calibration. The main table reports descriptive external maxima and ties within .001, whereas the paired analysis below separately evaluates uncertainty. Full-precision two- and three-rater statistics are supplied as aggregate CSV files.

Table 1: All families and conditions ($N = 305$); marginal 95% CIs.

**Accuracy**

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Human consensus | Spearman 0.809 [0.769, 0.842] | | | | | | QWK 0.704 [0.663, 0.743] | | | |
| Azure | Spearman 0.761 [0.709, 0.804] | | | | | | QWK 0.161 [0.129, 0.196] | | | |
| Family | S | | D | | H | | B-S | | B-D | |
| | $\rho$ | QWK | $\rho$ | QWK | $\rho$ | QWK | $\rho$ | QWK | $\rho$ | QWK |
| | 0.856 | 0.502 | 0.856 | 0.502 | 0.844 | 0.546 | 0.844 | 0.546 | 0.844 | 0.546 |
| speechocean | [0.821,0.882] | [0.451,0.549] | [0.821,0.882] | [0.451,0.549] | [0.807,0.872] | [0.494,0.592] | [0.807,0.872] | [0.494,0.592] | [0.807,0.872] | [0.494,0.592] |
| | 0.856 | 0.702 | 0.856 | 0.702 | 0.853 | 0.672 | 0.855 | 0.692 | 0.853 | 0.658 |
| korean | [0.822,0.882] | [0.662,0.739] | [0.822,0.882] | [0.662,0.739] | [0.820,0.881] | [0.631,0.711] | [0.822,0.882] | [0.652,0.730] | [0.819,0.880] | [0.617,0.697] |
| | 0.852 | 0.241 | 0.864 | 0.239 | 0.863 | 0.239 | 0.849 | 0.238 | 0.862 | 0.243 |
| aesop | [0.818,0.878] | [0.215,0.265] | [0.832,0.888] | [0.214,0.264] | [0.830,0.887] | [0.214,0.264] | [0.814,0.875] | [0.212,0.263] | [0.830,0.886] | [0.217,0.267] |
| | 0.853 | 0.821 | 0.841 | 0.798 | 0.849 | 0.819 | 0.852 | 0.821 | 0.841 | 0.802 |
| koreanocean | [0.816,0.882] | [0.789,0.847] | [0.802,0.871] | [0.764,0.827] | [0.811,0.879] | [0.788,0.846] | [0.813,0.881] | [0.789,0.847] | [0.803,0.872] | [0.769,0.831] |
| | 0.872 | 0.598 | 0.865 | 0.628 | 0.872 | 0.607 | 0.871 | 0.596 | 0.871 | 0.612 |
| aesopocean | [0.842,0.894] | [0.563,0.631] | [0.833,0.890] | [0.593,0.662] | [0.841,0.894] | [0.572,0.640] | [0.841,0.894] | [0.561,0.629] | [0.840,0.894] | [0.576,0.645] |
| | 0.867 | 0.477 | 0.867 | 0.477 | 0.867 | 0.472 | 0.867 | 0.477 | 0.867 | 0.475 |
| koreanaesop | [0.835,0.891] | [0.439,0.512] | [0.835,0.891] | [0.439,0.512] | [0.836,0.891] | [0.435,0.508] | [0.836,0.891] | [0.440,0.513] | [0.836,0.891] | [0.437,0.510] |
| | 0.870 | 0.585 | 0.862 | 0.576 | 0.857 | 0.575 | 0.869 | 0.579 | 0.863 | 0.577 |
| aesopkoreanocean | [0.839,0.894] | [0.548,0.620] | [0.829,0.888] | [0.537,0.613] | [0.824,0.884] | [0.537,0.611] | [0.837,0.893] | [0.540,0.614] | [0.830,0.889] | [0.538,0.613] |

**Fluency**

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Human consensus | Spearman 0.807 [0.768, 0.840] | | | | | | QWK 0.739 [0.695, 0.777] | | | |
| Azure | Spearman 0.633 [0.556, 0.698] | | | | | | QWK 0.078 [0.056, 0.107] | | | |
| Family | S | | D | | H | | B-S | | B-D | |
| | $\rho$ | QWK | $\rho$ | QWK | $\rho$ | QWK | $\rho$ | QWK | $\rho$ | QWK |
| | 0.811 | 0.716 | 0.847 | 0.629 | 0.842 | 0.670 | 0.832 | 0.704 | 0.836 | 0.693 |
| speechocean | [0.763,0.848] | [0.672,0.752] | [0.806,0.878] | [0.585,0.666] | [0.801,0.874] | [0.627,0.706] | [0.790,0.866] | [0.662,0.739] | [0.795,0.870] | [0.651,0.728] |
| | 0.860 | 0.495 | 0.860 | 0.495 | 0.859 | 0.501 | 0.864 | 0.536 | 0.859 | 0.503 |
| korean | [0.825,0.888] | [0.452,0.535] | [0.825,0.888] | [0.452,0.535] | [0.825,0.886] | [0.458,0.538] | [0.831,0.890] | [0.493,0.576] | [0.824,0.886] | [0.461,0.540] |
| | 0.850 | 0.276 | 0.855 | 0.312 | 0.859 | 0.315 | 0.855 | 0.289 | 0.859 | 0.314 |
| aesop | [0.813,0.879] | [0.247,0.305] | [0.819,0.883] | [0.279,0.342] | [0.823,0.885] | [0.282,0.346] | [0.819,0.883] | [0.258,0.318] | [0.823,0.886] | [0.281,0.345] |
| | 0.840 | 0.585 | 0.846 | 0.714 | 0.838 | 0.744 | 0.845 | 0.634 | 0.847 | 0.700 |
| koreanocean | [0.799,0.873] | [0.537,0.628] | [0.808,0.877] | [0.672,0.751] | [0.799,0.870] | [0.702,0.779] | [0.805,0.877] | [0.588,0.676] | [0.809,0.879] | [0.657,0.738] |
| | 0.874 | 0.613 | 0.874 | 0.613 | 0.862 | 0.612 | 0.883 | 0.751 | 0.881 | 0.742 |
| aesopocean | [0.842,0.899] | [0.575,0.647] | [0.842,0.899] | [0.575,0.647] | [0.828,0.890] | [0.574,0.647] | [0.854,0.906] | [0.718,0.779] | [0.851,0.905] | [0.710,0.771] |
| | 0.876 | 0.462 | 0.876 | 0.462 | 0.878 | 0.469 | 0.874 | 0.452 | 0.878 | 0.488 |
| koreanaesop | [0.846,0.900] | [0.424,0.496] | [0.846,0.900] | [0.424,0.496] | [0.848,0.901] | [0.431,0.502] | [0.843,0.898] | [0.415,0.486] | [0.848,0.902] | [0.451,0.522] |
| | 0.874 | 0.527 | 0.877 | 0.577 | 0.871 | 0.591 | 0.880 | 0.629 | 0.877 | 0.577 |
| aesopkoreanocean | [0.842,0.898] | [0.486,0.564] | [0.846,0.902] | [0.539,0.612] | [0.839,0.897] | [0.552,0.626] | [0.849,0.903] | [0.589,0.664] | [0.846,0.902] | [0.539,0.612] |

**Prosody**

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Human consensus | Spearman 0.807 [0.768, 0.842] | | | | | | QWK 0.742 [0.701, 0.778] | | | |
| Azure | Spearman 0.737 [0.672, 0.792] | | | | | | QWK 0.370 [0.318, 0.421] | | | |
| Family | S | | D | | H | | B-S | | B-D | |
| | $\rho$ | QWK | $\rho$ | QWK | $\rho$ | QWK | $\rho$ | QWK | $\rho$ | QWK |
| | 0.819 | 0.597 | 0.850 | 0.545 | 0.847 | 0.570 | 0.839 | 0.588 | 0.843 | 0.581 |
| speechocean | [0.776,0.855] | [0.551,0.639] | [0.813,0.880] | [0.501,0.585] | [0.809,0.878] | [0.525,0.610] | [0.800,0.871] | [0.543,0.628] | [0.804,0.874] | [0.535,0.622] |
| | 0.874 | 0.489 | 0.874 | 0.489 | 0.872 | 0.493 | 0.878 | 0.530 | 0.871 | 0.495 |
| korean | [0.842,0.899] | [0.447,0.526] | [0.842,0.899] | [0.447,0.526] | [0.840,0.896] | [0.452,0.531] | [0.847,0.902] | [0.487,0.568] | [0.839,0.895] | [0.454,0.533] |
| | 0.850 | 0.261 | 0.855 | 0.280 | 0.858 | 0.283 | 0.846 | 0.255 | 0.863 | 0.270 |
| aesop | [0.814,0.878] | [0.232,0.288] | [0.820,0.882] | [0.249,0.309] | [0.824,0.884] | [0.252,0.312] | [0.810,0.875] | [0.226,0.282] | [0.830,0.889] | [0.240,0.298] |
| | 0.850 | 0.546 | 0.851 | 0.641 | 0.840 | 0.675 | 0.856 | 0.584 | 0.854 | 0.629 |
| koreanocean | [0.812,0.881] | [0.501,0.586] | [0.815,0.882] | [0.599,0.680] | [0.802,0.872] | [0.632,0.712] | [0.818,0.886] | [0.540,0.624] | [0.817,0.885] | [0.586,0.668] |
| | 0.881 | 0.541 | 0.870 | 0.497 | 0.861 | 0.481 | 0.865 | 0.475 | 0.882 | 0.604 |
| aesopocean | [0.850,0.905] | [0.502,0.576] | [0.838,0.895] | [0.457,0.531] | [0.827,0.888] | [0.440,0.516] | [0.831,0.892] | [0.434,0.509] | [0.852,0.905] | [0.566,0.637] |
| | 0.883 | 0.431 | 0.883 | 0.431 | 0.883 | 0.433 | 0.883 | 0.431 | 0.889 | 0.446 |
| koreanaesop | [0.852,0.906] | [0.394,0.465] | [0.852,0.906] | [0.394,0.465] | [0.853,0.906] | [0.397,0.468] | [0.852,0.906] | [0.394,0.465] | [0.860,0.911] | [0.409,0.479] |
| | 0.881 | 0.480 | 0.879 | 0.505 | 0.872 | 0.518 | 0.882 | 0.493 | 0.879 | 0.505 |
| aesopkoreanocean | [0.850,0.906] | [0.440,0.516] | [0.847,0.904] | [0.467,0.541] | [0.839,0.898] | [0.479,0.554] | [0.850,0.906] | [0.454,0.529] | [0.847,0.904] | [0.467,0.541] |

# B Paired differences and score distributions

The same 305 recordings were jointly resampled 1,200,000 times per seed (42 and 2026), treating each unique recording URL as one speaker. Each replicate recomputed agreement against the same three two-rater reference pairs. Bonferroni intervals use percentile endpoints $0.025/510$ and $1 - 0.025/510$, covering 35 models versus human consensus and Azure, plus Azure versus human consensus, for three constructs and two metrics, and 84 banded-versus-previous-leader contrasts. Seven of 510 adjusted decisions differ between seeds and are marked with a dagger; these borderline comparisons do not establish a direction. The table reports seed-42 intervals; full-precision results for both seeds are released. Approximately 59 draws fall in each adjusted tail, so endpoints remain Monte Carlo approximations. Correction covers these contrasts, not the preceding model search or all historical experiments; inference conditions on the observed panel and fitted models. No private scores were used for fitting or calibration.

Table 2: Paired differences for all S/D/H/B configurations and constructs.

| | | | Accuracy | | | Fluency | | | Prosody | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| System | Reference | Metric | Diff. | 95% CI | Adjusted CI | Diff. | 95% CI | Adjusted CI | Diff. | 95% CI | Adjusted CI |
| **Azure benchmark (independent of selection rule)** | | | | | | | | | | | |
| azure | Human | $\rho$ | -0.049 | [-0.096, -0.005] | [-0.148, 0.038] | -0.174 | [-0.243, -0.110] | [-0.319, -0.052] | -0.070 | [-0.125, -0.022] | [-0.186, 0.019] |
| azure | Human | QWK | -0.543 | [-0.580, -0.504] | [-0.613, -0.464] | -0.661 | [-0.697, -0.620] | [-0.730, -0.578] | -0.372 | [-0.419, -0.325] | [-0.463, -0.277] |
| **S: 1-grid construct-wise correlation** | | | | | | | | | | | |
| speechocean | Human | $\rho$ | +0.046 | [0.016, 0.076] | [-0.016, 0.109] | +0.004 | [-0.036, 0.041] | [-0.080, 0.076] | +0.012 | [-0.023, 0.044] | [-0.061, 0.076] |
| speechocean | Human | QWK | -0.202 | [-0.249, -0.155] | [-0.294, -0.106] | -0.023 | [-0.067, 0.022] | [-0.112, 0.066] | -0.144 | [-0.188, -0.101] | [-0.231, -0.057] |
| speechocean | Azure | $\rho$ | +0.095 | [0.054, 0.139] | [0.015, 0.189] | +0.178 | [0.113, 0.247] | [0.052, 0.323] | +0.082 | [0.033, 0.135] | [-0.011, 0.197] |
| speechocean | Azure | QWK | +0.341 | [0.299, 0.381] | [0.258, 0.419] | +0.638 | [0.601, 0.671] | [0.561, 0.702] | +0.228 | [0.173, 0.282] | [0.117, 0.334] |
| korean | Human | $\rho$ | +0.046 | [0.018, 0.075] | [-0.012, 0.108] | +0.054 | [0.022, 0.085] | [-0.011, 0.120] | +0.067 | [0.039, 0.096] | [0.011, 0.128] |
| korean | Human | QWK | -0.002 | [-0.041, 0.037] | [-0.080, 0.076] | -0.244 | [-0.281, -0.208] | [-0.316, -0.171] | -0.253 | [-0.287, -0.219] | [-0.319, -0.187] |
| korean | Azure | $\rho$ | +0.095 | [0.055, 0.139] | [0.017, 0.189] | +0.228 | [0.164, 0.297] | [0.107, 0.373] | +0.137 | [0.098, 0.183] | [0.066, 0.237] |
| korean | Azure | QWK | +0.541 | [0.500, 0.580] | [0.460, 0.617] | +0.417 | [0.379, 0.452] | [0.341, 0.486] | +0.119 | [0.077, 0.160] | [0.034, 0.199] |
| aesop | Human | $\rho$ | +0.043 | [0.011, 0.074] | [-0.021, 0.110] | +0.044 | [0.009, 0.077] | [-0.026, 0.112] | +0.043 | [0.012, 0.073] | [-0.020, 0.106] |
| aesop | Human | QWK | -0.463 | [-0.493, -0.431] | [-0.521, -0.396] | -0.463 | [-0.494, -0.430] | [-0.524, -0.396] | -0.481 | [-0.510, -0.451] | [-0.536, -0.420] |
| aesop | Azure | $\rho$ | +0.092 | [0.054, 0.132] | [0.020, 0.179] | +0.218 | [0.157, 0.284] | [0.104, 0.358] | +0.113 | [0.073, 0.159] | [0.040, 0.213] |
| aesop | Azure | QWK | +0.080 | [0.053, 0.105] | [0.023, 0.128] | +0.198 | [0.174, 0.221] | [0.150, 0.242] | -0.109 | [-0.147, -0.072] | [-0.186, -0.037] |
| koreanocean | Human | $\rho$ | +0.044 | [0.013, 0.074] | [-0.021, 0.108] | +0.033 | [-0.004, 0.068] | [-0.045, 0.104] | +0.042 | [0.010, 0.074] | [-0.026, 0.106] |
| koreanocean | Human | QWK | +0.116 | [0.082, 0.152] | [0.048, 0.191] | -0.154 | [-0.195, -0.114] | [-0.236, -0.076] | -0.196 | [-0.232, -0.161] | [-0.268, -0.126] |
| koreanocean | Azure | $\rho$ | +0.093 | [0.051, 0.137] | [0.012, 0.187] | +0.207 | [0.140, 0.278] | [0.079, 0.359] | +0.113 | [0.069, 0.163] | [0.032, 0.222] |
| koreanocean | Azure | QWK | +0.660 | [0.626, 0.690] | [0.591, 0.718] | +0.507 | [0.464, 0.546] | [0.419, 0.584] | +0.176 | [0.128, 0.223] | [0.077, 0.269] |
| aesopocean | Human | $\rho$ | +0.062 | [0.033, 0.093] | [0.003, 0.127] | +0.068 | [0.038, 0.098] | [0.008, 0.131] | +0.073 | [0.048, 0.100] | [0.024, 0.129] |
| aesopocean | Human | QWK | -0.106 | [-0.140, -0.071] | [-0.172, -0.035] | -0.126 | [-0.159, -0.093] | [-0.192, -0.059] | -0.201 | [-0.232, -0.170] | [-0.262, -0.138] |
| aesopocean | Azure | $\rho$ | +0.111 | [0.071, 0.155] | [0.034, 0.206] | +0.242 | [0.180, 0.309] | [0.126, 0.385] | +0.144 | [0.099, 0.195] | [0.063, 0.256] |
| aesopocean | Azure | QWK | +0.437 | [0.402, 0.470] | [0.368, 0.503] | +0.535 | [0.501, 0.566] | [0.465, 0.595] | +0.171 | [0.128, 0.214] | [0.084, 0.254] |
| koreanaesop | Human | $\rho$ | +0.057 | [0.028, 0.088] | [-0.003, 0.121] | +0.069 | [0.040, 0.099] | [0.013, 0.133] | +0.076 | [0.049, 0.104] | [0.023, 0.135] |
| koreanaesop | Human | QWK | -0.227 | [-0.261, -0.193] | [-0.293, -0.157] | -0.277 | [-0.311, -0.243] | [-0.344, -0.210] | -0.311 | [-0.342, -0.279] | [-0.372, -0.247] |
| koreanaesop | Azure | $\rho$ | +0.106 | [0.066, 0.150] | [0.028, 0.201] | +0.243 | [0.182, 0.311] | [0.128, 0.385] | +0.146 | [0.105, 0.194] | [0.073, 0.250] |
| koreanaesop | Azure | QWK | +0.316 | [0.281, 0.350] | [0.247, 0.383] | +0.384 | [0.350, 0.415] | [0.317, 0.447] | +0.062 | [0.022, 0.100] | [-0.018, 0.136] |
| aesopkoreanocean | Human | $\rho$ | +0.061 | [0.032, 0.091] | [0.002, 0.125] | +0.067 | [0.037, 0.098] | [0.007, 0.131] | +0.074 | [0.047, 0.101] | [0.020, 0.132] |
| aesopkoreanocean | Human | QWK | -0.119 | [-0.153, -0.084] | [-0.186, -0.048] | -0.212 | [-0.247, -0.178] | [-0.281, -0.145] | -0.262 | [-0.293, -0.231] | [-0.323, -0.200] |
| aesopkoreanocean | Azure | $\rho$ | +0.110 | [0.070, 0.154] | [0.033, 0.204] | +0.241 | [0.178, 0.311] | [0.121, 0.388] | +0.144 | [0.101, 0.194] | [0.067, 0.253] |
| aesopkoreanocean | Azure | QWK | +0.425 | [0.388, 0.459] | [0.353, 0.492] | +0.449 | [0.412, 0.482] | [0.375, 0.514] | +0.111 | [0.068, 0.151] | [0.025, 0.191] |
| **D: 1-grid pronunciation-accuracy error** | | | | | | | | | | | |
| speechocean | Human | $\rho$ | +0.046 | [0.016, 0.076] | [-0.016, 0.109] | +0.040 | [0.006, 0.073] | [-0.032, 0.106] | +0.043 | [0.012, 0.072] | [-0.021, 0.103] |
| speechocean | Human | QWK | -0.202 | [-0.249, -0.155] | [-0.294, -0.106] | -0.111 | [-0.156, -0.064] | [-0.200, -0.016] | -0.196 | [-0.240, -0.151] | [-0.281, -0.104] |
| speechocean | Azure | $\rho$ | +0.095 | [0.054, 0.139] | [0.015, 0.189] | +0.214 | [0.150, 0.283] | [0.092, 0.359] | +0.113 | [0.066, 0.166] | [0.025, 0.229] |
| speechocean | Azure | QWK | +0.341 | [0.299, 0.381] | [0.258, 0.419] | +0.550 | [0.511, 0.586] | [0.470, 0.620] | +0.176 | [0.119, 0.232] | [0.062, 0.287] |
| korean | Human | $\rho$ | +0.046 | [0.018, 0.075] | [-0.012, 0.108] | +0.054 | [0.022, 0.085] | [-0.011, 0.120] | +0.067 | [0.039, 0.096] | [0.011, 0.128] |
| korean | Human | QWK | -0.002 | [-0.041, 0.037] | [-0.080, 0.076] | -0.244 | [-0.281, -0.208] | [-0.316, -0.171] | -0.253 | [-0.287, -0.219] | [-0.319, -0.187] |
| korean | Azure | $\rho$ | +0.095 | [0.055, 0.139] | [0.017, 0.189] | +0.228 | [0.164, 0.297] | [0.107, 0.373] | +0.137 | [0.098, 0.183] | [0.066, 0.237] |

| | | | Accuracy | | | Fluency | | | Prosody | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| System | Reference | Metric | Diff. | 95% CI | Adjusted CI | Diff. | 95% CI | Adjusted CI | Diff. | 95% CI | Adjusted CI |
| korean | Azure | QWK | +0.541 | [0.500, 0.580] | [0.460, 0.617] | +0.417 | [0.379, 0.452] | [0.341, 0.486] | +0.119 | [0.077, 0.160] | [0.034, 0.199] |
| aesop | Human | $\rho$ | +0.054 | [0.025, 0.085] | [-0.006, 0.119] | +0.048 | [0.015, 0.081] | [-0.020, 0.117] | +0.048 | [0.018, 0.078] | [-0.012, 0.110] |
| aesop | Human | QWK | -0.465 | [-0.493, -0.434] | [-0.520, -0.399] | -0.428 | [-0.459, -0.395] | [-0.489, -0.362] | -0.462 | [-0.491, -0.432] | [-0.518, -0.401] |
| aesop | Azure | $\rho$ | +0.103 | [0.065, 0.146] | [0.029, 0.195] | +0.222 | [0.163, 0.288] | [0.109, 0.360] | +0.119 | [0.077, 0.166] | [0.042, 0.220] |
| aesop | Azure | QWK | +0.079 | [0.052, 0.103] | [0.022, 0.126] | +0.233 | [0.206, 0.259] | [0.179, 0.285] | -0.090 | [-0.127, -0.053] | [-0.166, -0.018] |
| koreanocean | Human | $\rho$ | +0.031 | [-0.000, 0.062] | [-0.035, 0.094] | +0.039 | [0.005, 0.072] | [-0.031, 0.106] | +0.044 | [0.014, 0.073] | [-0.019, 0.104] |
| koreanocean | Human | QWK | +0.093 | [0.059, 0.129] | [0.024, 0.166] | -0.025 | [-0.063, 0.012] | [-0.101, 0.049] | -0.101 | [-0.135, -0.067] | [-0.169, -0.034] |
| koreanocean | Azure | $\rho$ | +0.080 | [0.037, 0.125] | [-0.003, 0.176] | +0.213 | [0.148, 0.283] | [0.090, 0.362] | +0.114 | [0.071, 0.164] | [0.035, 0.222] |
| koreanocean | Azure | QWK | +0.637 | [0.600, 0.671] | [0.563, 0.702] | +0.636 | [0.595, 0.672] | [0.554, 0.705] | +0.272 | [0.225, 0.317] | [0.177, 0.361] |
| aesopocean | Human | $\rho$ | +0.056 | [0.026, 0.086] | [-0.005, 0.120] | +0.068 | [0.038, 0.098] | [0.008, 0.131] | +0.063 | [0.037, 0.089] | [0.010, 0.119] |
| aesopocean | Human | QWK | -0.076 | [-0.109, -0.041] | [-0.142, -0.004] | -0.126 | [-0.159, -0.093] | [-0.192, -0.059] | -0.245 | [-0.276, -0.214] | [-0.305, -0.183] |
| aesopocean | Azure | $\rho$ | +0.105 | [0.063, 0.150] | [0.024, 0.203] | +0.242 | [0.180, 0.309] | [0.126, 0.385] | +0.133 | [0.089, 0.184] | [0.052, 0.245] |
| aesopocean | Azure | QWK | +0.468 | [0.432, 0.502] | [0.396, 0.534] | +0.535 | [0.501, 0.566] | [0.465, 0.595] | +0.127 | [0.085, 0.168] | [0.042, 0.208] |
| koreanaesop | Human | $\rho$ | +0.057 | [0.028, 0.088] | [-0.003, 0.121] | +0.069 | [0.040, 0.099] | [0.013, 0.133] | +0.076 | [0.049, 0.104] | [0.023, 0.135] |
| koreanaesop | Human | QWK | -0.227 | [-0.261, -0.193] | [-0.293, -0.157] | -0.277 | [-0.311, -0.243] | [-0.344, -0.210] | -0.311 | [-0.342, -0.279] | [-0.372, -0.247] |
| koreanaesop | Azure | $\rho$ | +0.106 | [0.066, 0.150] | [0.028, 0.201] | +0.243 | [0.182, 0.311] | [0.128, 0.385] | +0.146 | [0.105, 0.194] | [0.073, 0.250] |
| koreanaesop | Azure | QWK | +0.316 | [0.281, 0.350] | [0.247, 0.383] | +0.384 | [0.350, 0.415] | [0.317, 0.447] | +0.062 | [0.022, 0.100] | [-0.018, 0.136] |
| aesopkoreanocean | Human | $\rho$ | +0.052 | [0.022, 0.083] | [-0.009, 0.117] | +0.070 | [0.040, 0.101] | [0.010, 0.135] | +0.072 | [0.045, 0.099] | [0.017, 0.130] |
| aesopkoreanocean | Human | QWK | -0.128 | [-0.162, -0.094] | [-0.195, -0.058] | -0.162 | [-0.196, -0.128] | [-0.230, -0.094] | -0.237 | [-0.268, -0.205] | [-0.298, -0.175] |
| aesopkoreanocean | Azure | $\rho$ | +0.101 | [0.060, 0.146] | [0.023, 0.198] | +0.244 | [0.182, 0.313] | [0.127, 0.390] | +0.142 | [0.099, 0.191] | [0.066, 0.249] |
| aesopkoreanocean | Azure | QWK | +0.415 | [0.378, 0.451] | [0.342, 0.485] | +0.499 | [0.463, 0.532] | [0.426, 0.563] | +0.136 | [0.095, 0.175] | [0.053, 0.213] |
| **H: 5-grid pronunciation-accuracy error** | | | | | | | | | | | |
| speechocean | Human | $\rho$ | +0.035 | [0.004, 0.065] | [-0.029, 0.097] | +0.035 | [-0.000, 0.069] | [-0.039, 0.102] | +0.040 | [0.009, 0.069] | [-0.024, 0.099] |
| speechocean | Human | QWK | -0.158 | [-0.206, -0.110] | [-0.253, -0.061] | -0.069 | [-0.113, -0.024] | [-0.156, 0.023] | -0.172 | [-0.215, -0.128] | [-0.255, -0.083] |
| speechocean | Azure | $\rho$ | +0.083 | [0.041, 0.128] | [-0.000, 0.178]$^{\dagger}$ | +0.209 | [0.145, 0.278] | [0.086, 0.354] | +0.110 | [0.063, 0.163] | [0.021, 0.225] |
| speechocean | Azure | QWK | +0.385 | [0.341, 0.426] | [0.298, 0.464] | +0.592 | [0.553, 0.627] | [0.513, 0.660] | +0.200 | [0.145, 0.255] | [0.088, 0.308] |
| korean | Human | $\rho$ | +0.044 | [0.015, 0.073] | [-0.015, 0.106] | +0.053 | [0.021, 0.084] | [-0.012, 0.119] | +0.064 | [0.036, 0.093] | [0.007, 0.127] |
| korean | Human | QWK | -0.032 | [-0.071, 0.008] | [-0.111, 0.047] | -0.239 | [-0.276, -0.202] | [-0.311, -0.166] | -0.249 | [-0.283, -0.214] | [-0.316, -0.181] |
| korean | Azure | $\rho$ | +0.093 | [0.052, 0.137] | [0.013, 0.187] | +0.227 | [0.163, 0.296] | [0.106, 0.372] | +0.135 | [0.094, 0.181] | [0.061, 0.236] |
| korean | Azure | QWK | +0.511 | [0.470, 0.551] | [0.428, 0.589] | +0.422 | [0.385, 0.457] | [0.347, 0.491] | +0.124 | [0.082, 0.164] | [0.039, 0.203] |
| aesop | Human | $\rho$ | +0.053 | [0.024, 0.083] | [-0.007, 0.118] | +0.052 | [0.019, 0.085] | [-0.015, 0.120] | +0.051 | [0.022, 0.080] | [-0.008, 0.112] |
| aesop | Human | QWK | -0.465 | [-0.493, -0.434] | [-0.520, -0.399] | -0.424 | [-0.455, -0.391] | [-0.485, -0.358] | -0.459 | [-0.488, -0.429] | [-0.515, -0.397] |
| aesop | Azure | $\rho$ | +0.102 | [0.063, 0.145] | [0.027, 0.193] | +0.226 | [0.166, 0.291] | [0.113, 0.364] | +0.122 | [0.080, 0.168] | [0.046, 0.223] |
| aesop | Azure | QWK | +0.079 | [0.052, 0.103] | [0.022, 0.126] | +0.237 | [0.209, 0.263] | [0.183, 0.288] | -0.087 | [-0.124, -0.050] | [-0.162, -0.016] |
| koreanocean | Human | $\rho$ | +0.040 | [0.009, 0.071] | [-0.026, 0.103] | +0.032 | [-0.002, 0.064] | [-0.039, 0.099] | +0.033 | [0.002, 0.062] | [-0.031, 0.093] |
| koreanocean | Human | QWK | +0.115 | [0.080, 0.151] | [0.045, 0.190] | +0.005 | [-0.033, 0.042] | [-0.070, 0.080] | -0.067 | [-0.102, -0.033] | [-0.137, 0.001] |
| koreanocean | Azure | $\rho$ | +0.089 | [0.047, 0.134] | [0.007, 0.184] | +0.206 | [0.141, 0.276] | [0.082, 0.354] | +0.103 | [0.059, 0.153] | [0.023, 0.210] |
| koreanocean | Azure | QWK | +0.659 | [0.624, 0.690] | [0.587, 0.719] | +0.666 | [0.627, 0.701] | [0.586, 0.733] | +0.305 | [0.258, 0.351] | [0.210, 0.395] |
| aesopocean | Human | $\rho$ | +0.062 | [0.033, 0.092] | [0.003, 0.126] | +0.056 | [0.024, 0.087] | [-0.008, 0.121] | +0.054 | [0.026, 0.081] | [-0.003, 0.111] |
| aesopocean | Human | QWK | -0.097 | [-0.131, -0.063] | [-0.163, -0.027] | -0.127 | [-0.161, -0.093] | [-0.194, -0.059] | -0.261 | [-0.292, -0.230] | [-0.322, -0.199] |
| aesopocean | Azure | $\rho$ | +0.111 | [0.070, 0.156] | [0.033, 0.208] | +0.230 | [0.168, 0.297] | [0.114, 0.372] | +0.124 | [0.079, 0.175] | [0.041, 0.236] |
| aesopocean | Azure | QWK | +0.446 | [0.411, 0.480] | [0.376, 0.512] | +0.534 | [0.500, 0.565] | [0.464, 0.595] | +0.111 | [0.068, 0.152] | [0.024, 0.192] |
| koreanaesop | Human | $\rho$ | +0.058 | [0.029, 0.088] | [-0.001, 0.121]$^{\dagger}$ | +0.071 | [0.042, 0.101] | [0.014, 0.135] | +0.076 | [0.049, 0.104] | [0.023, 0.136] |
| koreanaesop | Human | QWK | -0.232 | [-0.265, -0.198] | [-0.296, -0.163] | -0.271 | [-0.304, -0.237] | [-0.337, -0.202] | -0.308 | [-0.340, -0.277] | [-0.370, -0.245] |
| koreanaesop | Azure | $\rho$ | +0.107 | [0.067, 0.151] | [0.029, 0.202] | +0.245 | [0.183, 0.312] | [0.129, 0.388] | +0.147 | [0.105, 0.194] | [0.072, 0.250] |
| koreanaesop | Azure | QWK | +0.312 | [0.277, 0.345] | [0.243, 0.378] | +0.390 | [0.356, 0.422] | [0.323, 0.453] | +0.064 | [0.025, 0.102] | [-0.015, 0.138] |
| aesopkoreanocean | Human | $\rho$ | +0.048 | [0.018, 0.078] | [-0.014, 0.112] | +0.065 | [0.034, 0.096] | [0.004, 0.130] | +0.065 | [0.038, 0.092] | [0.009, 0.123] |
| aesopkoreanocean | Human | QWK | -0.129 | [-0.163, -0.094] | [-0.196, -0.059] | -0.148 | [-0.183, -0.113] | [-0.217, -0.078] | -0.224 | [-0.256, -0.192] | [-0.288, -0.160] |
| aesopkoreanocean | Azure | $\rho$ | +0.097 | [0.056, 0.141] | [0.018, 0.193] | +0.239 | [0.176, 0.307] | [0.121, 0.384] | +0.135 | [0.093, 0.184] | [0.058, 0.242] |
| aesopkoreanocean | Azure | QWK | +0.415 | [0.377, 0.450] | [0.341, 0.485] | +0.513 | [0.477, 0.546] | [0.440, 0.578] | +0.148 | [0.107, 0.188] | [0.064, 0.227] |
| **B-S: three-ratio banded, construct-wise correlation** | | | | | | | | | | | |
| speechocean | Human | $\rho$ | +0.035 | [0.004, 0.065] | [-0.029, 0.097] | +0.025 | [-0.011, 0.060] | [-0.052, 0.093] | +0.032 | [0.000, 0.062] | [-0.035, 0.092] |
| speechocean | Human | QWK | -0.158 | [-0.206, -0.110] | [-0.253, -0.061] | -0.035 | [-0.079, 0.009] | [-0.123, 0.053] | -0.154 | [-0.196, -0.111] | [-0.237, -0.067] |

| | | | Accuracy | | | Fluency | | | Prosody | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| System | Reference | Metric | Diff. | 95% CI | Adjusted CI | Diff. | 95% CI | Adjusted CI | Diff. | 95% CI | Adjusted CI |
| speechocean | Azure | $\rho$ | +0.083 | [0.041, 0.128] | [-0.000, 0.178]† | +0.199 | [0.135, 0.268] | [0.076, 0.345] | +0.102 | [0.054, 0.155] | [0.012, 0.217] |
| speechocean | Azure | QWK | +0.385 | [0.341, 0.426] | [0.298, 0.464] | +0.626 | [0.589, 0.659] | [0.549, 0.691] | +0.218 | [0.163, 0.273] | [0.108, 0.325] |
| korean | Human | $\rho$ | +0.046 | [0.017, 0.075] | [-0.012, 0.108] | +0.058 | [0.027, 0.088] | [-0.003, 0.122] | +0.071 | [0.044, 0.098] | [0.017, 0.130] |
| korean | Human | QWK | -0.012 | [-0.051, 0.027] | [-0.090, 0.067] | -0.203 | [-0.241, -0.166] | [-0.278, -0.130] | -0.211 | [-0.245, -0.178] | [-0.279, -0.145] |
| korean | Azure | $\rho$ | +0.095 | [0.054, 0.139] | [0.016, 0.189] | +0.232 | [0.169, 0.300] | [0.112, 0.377] | +0.141 | [0.100, 0.189] | [0.068, 0.246] |
| korean | Azure | QWK | +0.531 | [0.490, 0.571] | [0.450, 0.608] | +0.458 | [0.419, 0.493] | [0.380, 0.527] | +0.161 | [0.117, 0.203] | [0.072, 0.245] |
| aesop | Human | $\rho$ | +0.039 | [0.008, 0.071] | [-0.024, 0.106] | +0.049 | [0.015, 0.081] | [-0.019, 0.116] | +0.039 | [0.008, 0.070] | [-0.025, 0.103] |
| aesop | Human | QWK | -0.466 | [-0.496, -0.434] | [-0.523, -0.398] | -0.451 | [-0.482, -0.418] | [-0.512, -0.384] | -0.487 | [-0.515, -0.456] | [-0.542, -0.424] |
| aesop | Azure | $\rho$ | +0.088 | [0.051, 0.128] | [0.017, 0.174] | +0.223 | [0.162, 0.289] | [0.109, 0.363] | +0.110 | [0.069, 0.155] | [0.037, 0.209] |
| aesop | Azure | QWK | +0.077 | [0.051, 0.102] | [0.021, 0.125] | +0.210 | [0.185, 0.234] | [0.160, 0.257] | -0.115 | [-0.153, -0.077] | [-0.192, -0.042] |
| koreanocean | Human | $\rho$ | +0.042 | [0.011, 0.073] | [-0.023, 0.106] | +0.038 | [0.002, 0.073] | [-0.038, 0.109] | +0.048 | [0.017, 0.078] | [-0.017, 0.110] |
| koreanocean | Human | QWK | +0.116 | [0.082, 0.152] | [0.047, 0.191] | -0.105 | [-0.144, -0.066] | [-0.185, -0.028] | -0.158 | [-0.192, -0.124] | [-0.226, -0.091] |
| koreanocean | Azure | $\rho$ | +0.091 | [0.049, 0.136] | [0.010, 0.187] | +0.212 | [0.146, 0.284] | [0.086, 0.364] | +0.119 | [0.075, 0.168] | [0.039, 0.227] |
| koreanocean | Azure | QWK | +0.660 | [0.626, 0.691] | [0.589, 0.719] | +0.556 | [0.513, 0.595] | [0.468, 0.631] | +0.215 | [0.168, 0.260] | [0.119, 0.304] |
| aesopocean | Human | $\rho$ | +0.062 | [0.032, 0.092] | [0.002, 0.127] | +0.077 | [0.048, 0.107] | [0.020, 0.140] | +0.058 | [0.031, 0.085] | [0.004, 0.114] |
| aesopocean | Human | QWK | -0.109 | [-0.142, -0.074] | [-0.175, -0.038] | +0.012 | [-0.022, 0.046] | [-0.055, 0.082] | -0.267 | [-0.298, -0.237] | [-0.327, -0.205] |
| aesopocean | Azure | $\rho$ | +0.111 | [0.071, 0.155] | [0.034, 0.206] | +0.251 | [0.188, 0.319] | [0.134, 0.396] | +0.128 | [0.084, 0.178] | [0.047, 0.239] |
| aesopocean | Azure | QWK | +0.435 | [0.400, 0.468] | [0.366, 0.500] | +0.673 | [0.640, 0.702] | [0.605, 0.729] | +0.105 | [0.063, 0.146] | [0.019, 0.185] |
| koreanaesop | Human | $\rho$ | +0.058 | [0.028, 0.088] | [-0.002, 0.122] | +0.068 | [0.038, 0.097] | [0.010, 0.131] | +0.076 | [0.049, 0.104] | [0.023, 0.135] |
| koreanaesop | Human | QWK | -0.227 | [-0.261, -0.192] | [-0.292, -0.157] | -0.287 | [-0.321, -0.253] | [-0.353, -0.219] | -0.311 | [-0.342, -0.279] | [-0.372, -0.247] |
| koreanaesop | Azure | $\rho$ | +0.107 | [0.066, 0.150] | [0.029, 0.201] | +0.242 | [0.180, 0.309] | [0.126, 0.384] | +0.146 | [0.105, 0.194] | [0.073, 0.250] |
| koreanaesop | Azure | QWK | +0.316 | [0.281, 0.350] | [0.247, 0.383] | +0.374 | [0.340, 0.405] | [0.307, 0.436] | +0.062 | [0.022, 0.100] | [-0.018, 0.136] |
| aesopkoreanocean | Human | $\rho$ | +0.059 | [0.030, 0.090] | [-0.001, 0.123]† | +0.073 | [0.043, 0.104] | [0.013, 0.138] | +0.075 | [0.048, 0.102] | [0.021, 0.133] |
| aesopkoreanocean | Human | QWK | -0.125 | [-0.160, -0.091] | [-0.193, -0.055] | -0.111 | [-0.145, -0.076] | [-0.180, -0.042] | -0.249 | [-0.279, -0.218] | [-0.310, -0.187] |
| aesopkoreanocean | Azure | $\rho$ | +0.108 | [0.068, 0.152] | [0.031, 0.204] | +0.247 | [0.184, 0.316] | [0.129, 0.394] | +0.145 | [0.103, 0.195] | [0.069, 0.253] |
| aesopkoreanocean | Azure | QWK | +0.418 | [0.381, 0.453] | [0.345, 0.486] | +0.551 | [0.514, 0.584] | [0.476, 0.616] | +0.124 | [0.083, 0.164] | [0.040, 0.202] |
| **B-D: three-ratio banded, pronunciation-accuracy error** | | | | | | | | | | | |
| speechocean | Human | $\rho$ | +0.035 | [0.004, 0.065] | [-0.029, 0.097] | +0.030 | [-0.006, 0.064] | [-0.045, 0.097] | +0.035 | [0.004, 0.065] | [-0.030, 0.095] |
| speechocean | Human | QWK | -0.158 | [-0.206, -0.110] | [-0.253, -0.061] | -0.046 | [-0.090, -0.001] | [-0.133, 0.044] | -0.161 | [-0.203, -0.117] | [-0.244, -0.073] |
| speechocean | Azure | $\rho$ | +0.083 | [0.041, 0.128] | [-0.000, 0.178]† | +0.204 | [0.139, 0.273] | [0.080, 0.349] | +0.106 | [0.058, 0.159] | [0.016, 0.221] |
| speechocean | Azure | QWK | +0.385 | [0.341, 0.426] | [0.298, 0.464] | +0.615 | [0.577, 0.649] | [0.537, 0.681] | +0.212 | [0.156, 0.266] | [0.101, 0.319] |
| korean | Human | $\rho$ | +0.043 | [0.014, 0.072] | [-0.016, 0.106] | +0.052 | [0.020, 0.084] | [-0.012, 0.118] | +0.063 | [0.035, 0.093] | [0.006, 0.126] |
| korean | Human | QWK | -0.046 | [-0.086, -0.007] | [-0.125, 0.032] | -0.236 | [-0.273, -0.200] | [-0.309, -0.163] | -0.246 | [-0.281, -0.212] | [-0.314, -0.178] |
| korean | Azure | $\rho$ | +0.092 | [0.051, 0.136] | [0.012, 0.186] | +0.226 | [0.162, 0.295] | [0.106, 0.371] | +0.134 | [0.093, 0.181] | [0.060, 0.235] |
| korean | Azure | QWK | +0.497 | [0.455, 0.537] | [0.415, 0.575] | +0.425 | [0.387, 0.460] | [0.350, 0.494] | +0.126 | [0.084, 0.166] | [0.041, 0.205] |
| aesop | Human | $\rho$ | +0.052 | [0.022, 0.083] | [-0.008, 0.118] | +0.052 | [0.019, 0.085] | [-0.015, 0.119] | +0.056 | [0.027, 0.085] | [-0.002, 0.116] |
| aesop | Human | QWK | -0.462 | [-0.490, -0.430] | [-0.518, -0.396] | -0.425 | [-0.457, -0.392] | [-0.487, -0.359] | -0.472 | [-0.500, -0.442] | [-0.526, -0.410] |
| aesop | Azure | $\rho$ | +0.101 | [0.063, 0.143] | [0.028, 0.191] | +0.226 | [0.165, 0.292] | [0.112, 0.367] | +0.126 | [0.084, 0.174] | [0.049, 0.230] |
| aesop | Azure | QWK | +0.082 | [0.055, 0.107] | [0.025, 0.130] | +0.236 | [0.208, 0.263] | [0.181, 0.289] | -0.099 | [-0.137, -0.062] | [-0.176, -0.027] |
| koreanocean | Human | $\rho$ | +0.032 | [-0.000, 0.062] | [-0.035, 0.095] | +0.041 | [0.006, 0.074] | [-0.030, 0.109] | +0.047 | [0.017, 0.076] | [-0.016, 0.108] |
| koreanocean | Human | QWK | +0.098 | [0.063, 0.134] | [0.028, 0.170] | -0.039 | [-0.077, -0.002] | [-0.116, 0.035] | -0.113 | [-0.147, -0.080] | [-0.181, -0.047] |
| koreanocean | Azure | $\rho$ | +0.081 | [0.038, 0.126] | [-0.003, 0.177] | +0.215 | [0.150, 0.285] | [0.092, 0.364] | +0.117 | [0.074, 0.167] | [0.039, 0.224] |
| koreanocean | Azure | QWK | +0.641 | [0.605, 0.675] | [0.568, 0.706] | +0.622 | [0.581, 0.658] | [0.538, 0.692] | +0.259 | [0.212, 0.304] | [0.164, 0.349] |
| aesopocean | Human | $\rho$ | +0.061 | [0.032, 0.092] | [0.002, 0.126] | +0.075 | [0.045, 0.105] | [0.016, 0.139] | +0.074 | [0.049, 0.101] | [0.024, 0.131] |
| aesopocean | Human | QWK | -0.092 | [-0.126, -0.058] | [-0.158, -0.022] | +0.003 | [-0.030, 0.037] | [-0.062, 0.072] | -0.138 | [-0.169, -0.107] | [-0.199, -0.076] |
| aesopocean | Azure | $\rho$ | +0.110 | [0.069, 0.155] | [0.031, 0.207] | +0.249 | [0.187, 0.317] | [0.132, 0.393] | +0.145 | [0.100, 0.196] | [0.064, 0.256] |
| aesopocean | Azure | QWK | +0.451 | [0.416, 0.484] | [0.381, 0.517] | +0.664 | [0.632, 0.693] | [0.598, 0.721] | +0.234 | [0.191, 0.276] | [0.147, 0.318] |
| koreanaesop | Human | $\rho$ | +0.058 | [0.029, 0.088] | [-0.001, 0.121]† | +0.072 | [0.043, 0.102] | [0.015, 0.136] | +0.082 | [0.056, 0.109] | [0.031, 0.141] |
| koreanaesop | Human | QWK | -0.229 | [-0.263, -0.195] | [-0.294, -0.160] | -0.251 | [-0.285, -0.216] | [-0.319, -0.181] | -0.296 | [-0.327, -0.264] | [-0.357, -0.232] |
| koreanaesop | Azure | $\rho$ | +0.107 | [0.067, 0.151] | [0.029, 0.202] | +0.246 | [0.184, 0.313] | [0.130, 0.388] | +0.152 | [0.109, 0.202] | [0.075, 0.261] |
| koreanaesop | Azure | QWK | +0.314 | [0.279, 0.348] | [0.245, 0.380] | +0.410 | [0.376, 0.443] | [0.342, 0.475] | +0.077 | [0.037, 0.115] | [-0.003, 0.152] |
| aesopkoreanocean | Human | $\rho$ | +0.054 | [0.024, 0.084] | [-0.007, 0.118] | +0.070 | [0.040, 0.101] | [0.010, 0.135] | +0.072 | [0.045, 0.099] | [0.017, 0.130] |
| aesopkoreanocean | Human | QWK | -0.127 | [-0.161, -0.093] | [-0.194, -0.057] | -0.162 | [-0.196, -0.128] | [-0.230, -0.094] | -0.237 | [-0.268, -0.205] | [-0.298, -0.175] |
| aesopkoreanocean | Azure | $\rho$ | +0.103 | [0.062, 0.147] | [0.024, 0.199] | +0.244 | [0.182, 0.313] | [0.127, 0.390] | +0.142 | [0.099, 0.191] | [0.066, 0.249] |

| | | | Accuracy | | | Fluency | | | Prosody | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| System | Reference | Metric | Diff. | 95% CI | Adjusted CI | Diff. | 95% CI | Adjusted CI | Diff. | 95% CI | Adjusted CI |
| aesopkoreanocean | Azure | QWK | +0.416 | [0.379, 0.452] | [0.343, 0.486] | +0.499 | [0.463, 0.532] | [0.426, 0.563] | +0.136 | [0.095, 0.175] | [0.053, 0.213] |
| **Banded minus previous S/D/H metric leader** | | | | | | | | | | | |
| **B-S** | | | | | | | | | | | |
| speechocean | Previous best | $\rho$ | -0.028 | [-0.047, -0.010] | [-0.070, 0.007] | -0.046 | [-0.069, -0.025] | [-0.096, -0.006] | -0.044 | [-0.068, -0.023] | [-0.097, -0.005] |
| speechocean | Previous best | QWK | -0.275 | [-0.314, -0.236] | [-0.350, -0.196] | -0.040 | [-0.074, -0.006] | [-0.109, 0.029] | -0.087 | [-0.123, -0.049] | [-0.159, -0.011] |
| korean | Previous best | $\rho$ | -0.016 | [-0.031, -0.003] | [-0.047, 0.011] | -0.013 | [-0.022, -0.005] | [-0.034, 0.003] | -0.005 | [-0.014, 0.003] | [-0.023, 0.011] |
| korean | Previous best | QWK | -0.128 | [-0.152, -0.105] | [-0.175, -0.082] | -0.208 | [-0.230, -0.187] | [-0.252, -0.167] | -0.144 | [-0.164, -0.125] | [-0.184, -0.107] |
| aesop | Previous best | $\rho$ | -0.023 | [-0.036, -0.011] | [-0.052, 0.000]† | -0.022 | [-0.035, -0.011] | [-0.051, 0.000] | -0.037 | [-0.052, -0.024] | [-0.070, -0.012] |
| aesop | Previous best | QWK | -0.582 | [-0.600, -0.564] | [-0.617, -0.545] | -0.455 | [-0.479, -0.431] | [-0.501, -0.405] | -0.419 | [-0.443, -0.394] | [-0.466, -0.368] |
| koreanocean | Previous best | $\rho$ | -0.020 | [-0.039, -0.003] | [-0.061, 0.012] | -0.033 | [-0.056, -0.012] | [-0.084, 0.007] | -0.028 | [-0.049, -0.009] | [-0.074, 0.009] |
| koreanocean | Previous best | QWK | -0.000 | [-0.004, 0.004] | [-0.008, 0.008] | -0.110 | [-0.127, -0.093] | [-0.145, -0.077] | -0.090 | [-0.105, -0.076] | [-0.121, -0.062] |
| aesopocean | Previous best | $\rho$ | -0.000 | [-0.002, 0.001] | [-0.004, 0.003] | +0.006 | [-0.004, 0.016] | [-0.015, 0.028] | -0.018 | [-0.033, -0.005] | [-0.049, 0.009] |
| aesopocean | Previous best | QWK | -0.225 | [-0.246, -0.205] | [-0.266, -0.183] | +0.007 | [-0.013, 0.028] | [-0.033, 0.051] | -0.200 | [-0.222, -0.178] | [-0.244, -0.158] |
| koreanaesop | Previous best | $\rho$ | -0.005 | [-0.011, 0.002] | [-0.019, 0.008] | -0.003 | [-0.008, 0.001] | [-0.013, 0.006] | -0.000 | [-0.003, 0.003] | [-0.005, 0.006] |
| koreanaesop | Previous best | QWK | -0.343 | [-0.368, -0.318] | [-0.394, -0.294] | -0.292 | [-0.314, -0.270] | [-0.336, -0.249] | -0.243 | [-0.264, -0.223] | [-0.285, -0.204] |
| aesopkoreanocean | Previous best | $\rho$ | -0.003 | [-0.009, 0.002] | [-0.016, 0.008] | +0.002 | [-0.008, 0.013] | [-0.021, 0.024] | -0.001 | [-0.012, 0.009] | [-0.024, 0.020] |
| aesopkoreanocean | Previous best | QWK | -0.242 | [-0.266, -0.219] | [-0.289, -0.196] | -0.115 | [-0.132, -0.099] | [-0.150, -0.083] | -0.181 | [-0.197, -0.166] | [-0.214, -0.150] |
| **B-D** | | | | | | | | | | | |
| speechocean | Previous best | $\rho$ | -0.028 | [-0.047, -0.010] | [-0.070, 0.007] | -0.041 | [-0.064, -0.021] | [-0.091, -0.002] | -0.041 | [-0.063, -0.020] | [-0.091, -0.002] |
| speechocean | Previous best | QWK | -0.275 | [-0.314, -0.236] | [-0.350, -0.196] | -0.051 | [-0.085, -0.017] | [-0.119, 0.019] | -0.093 | [-0.130, -0.056] | [-0.165, -0.017] |
| korean | Previous best | $\rho$ | -0.019 | [-0.034, -0.005] | [-0.052, 0.009] | -0.019 | [-0.029, -0.010] | [-0.042, -0.002] | -0.013 | [-0.021, -0.005] | [-0.031, 0.002] |
| korean | Previous best | QWK | -0.163 | [-0.188, -0.137] | [-0.213, -0.113] | -0.241 | [-0.262, -0.220] | [-0.284, -0.201] | -0.179 | [-0.198, -0.161] | [-0.217, -0.144] |
| aesop | Previous best | $\rho$ | -0.010 | [-0.020, -0.000] | [-0.031, 0.010] | -0.019 | [-0.031, -0.008] | [-0.046, 0.001] | -0.020 | [-0.030, -0.011] | [-0.043, -0.002] |
| aesop | Previous best | QWK | -0.578 | [-0.596, -0.559] | [-0.615, -0.539] | -0.430 | [-0.453, -0.405] | [-0.475, -0.381] | -0.404 | [-0.427, -0.380] | [-0.450, -0.354] |
| koreanocean | Previous best | $\rho$ | -0.031 | [-0.050, -0.014] | [-0.072, 0.002] | -0.030 | [-0.050, -0.013] | [-0.074, 0.003] | -0.029 | [-0.048, -0.012] | [-0.070, 0.003] |
| koreanocean | Previous best | QWK | -0.019 | [-0.030, -0.007] | [-0.042, 0.004] | -0.044 | [-0.053, -0.036] | [-0.062, -0.028] | -0.046 | [-0.054, -0.038] | [-0.063, -0.030] |
| aesopocean | Previous best | $\rho$ | -0.001 | [-0.005, 0.003] | [-0.009, 0.007] | +0.004 | [-0.007, 0.014] | [-0.019, 0.026] | -0.002 | [-0.012, 0.009] | [-0.023, 0.021] |
| aesopocean | Previous best | QWK | -0.209 | [-0.231, -0.187] | [-0.254, -0.164] | -0.002 | [-0.021, 0.019] | [-0.040, 0.042] | -0.071 | [-0.092, -0.049] | [-0.114, -0.028] |
| koreanaesop | Previous best | $\rho$ | -0.004 | [-0.011, 0.003] | [-0.020, 0.010] | +0.001 | [-0.003, 0.005] | [-0.007, 0.009] | +0.006 | [0.002, 0.010] | [-0.001, 0.015] |
| koreanaesop | Previous best | QWK | -0.346 | [-0.371, -0.321] | [-0.396, -0.296] | -0.255 | [-0.278, -0.233] | [-0.301, -0.212] | -0.228 | [-0.250, -0.207] | [-0.271, -0.187] |
| aesopkoreanocean | Previous best | $\rho$ | -0.009 | [-0.016, -0.001] | [-0.026, 0.005] | -0.000 | [-0.011, 0.010] | [-0.023, 0.021] | -0.005 | [-0.015, 0.005] | [-0.027, 0.015] |
| aesopkoreanocean | Previous best | QWK | -0.243 | [-0.269, -0.219] | [-0.295, -0.195] | -0.167 | [-0.182, -0.152] | [-0.198, -0.138] | -0.169 | [-0.184, -0.155] | [-0.199, -0.142] |

*Note.* Positive differences favor the system; Human denotes human–other-rater consensus. Adjusted intervals cover 510 contrasts. Decisions use unrounded endpoints, not displayed zeros. QWK uses clipping and half-up rounding; Spearman retains raw scores. Previous accuracy leaders: S-aesopocean for correlation and S-koreanocean for QWK; fluency/prosody leaders: H-koreanaesop for correlation and H-koreanocean for QWK. A dagger marks seed-dependent interval exclusion.

### B.1 Score distributions

The ridge distributions below use the Spearman-selected (S) condition; Azure is independent of selection rule. Means, sample standard deviations, bias, and MAE use unrounded predictions on the 0–100 scale against the three-rater mean; no recalibration is fitted.

Table 3: Score distributions for the S condition and Azure.

| System | Accuracy | | | | Fluency | | | | Prosody | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Mean | SD | Bias | MAE | Mean | SD | Bias | MAE | Mean | SD | Bias | MAE |
| azure | 89.29 | 7.50 | 23.63 | 23.63 | 94.18 | 5.87 | 27.88 | 27.88 | 77.13 | 6.15 | 10.57 | 11.04 |
| speechocean | 73.57 | 6.14 | 7.90 | 8.96 | 64.67 | 7.37 | -1.63 | 5.26 | 60.79 | 6.81 | -5.76 | 6.70 |
| korean | 58.57 | 12.39 | -7.10 | 7.66 | 53.31 | 10.75 | -12.99 | 12.99 | 53.31 | 10.75 | -13.25 | 13.25 |
| aesop | 38.49 | 13.31 | -27.17 | 27.17 | 44.43 | 10.97 | -21.87 | 21.87 | 44.14 | 10.62 | -22.42 | 22.42 |
| koreanocean | 64.85 | 9.77 | -0.81 | 4.38 | 57.68 | 8.45 | -8.62 | 8.74 | 56.89 | 8.32 | -9.66 | 9.72 |
| aesopocean | 55.27 | 11.66 | -10.40 | 10.51 | 56.63 | 10.99 | -9.67 | 9.79 | 55.25 | 9.89 | -11.31 | 11.33 |
| koreanaesop | 50.09 | 14.23 | -15.58 | 15.59 | 51.09 | 12.44 | -15.21 | 15.21 | 50.26 | 12.36 | -16.29 | 16.29 |
| aesopkoreanocean | 54.37 | 12.52 | -11.30 | 11.38 | 54.18 | 10.50 | -12.12 | 12.13 | 53.05 | 10.42 | -13.50 | 13.50 |

## C Public-development comparisons

The matched seven-family experiment uses fixed representation combinations and Spearman-based penalty selection (S) over included-corpus development sets. All entries are development Spearman correlations, not final-test results. SpeechOcean uses the matched fitting procedure, not the original historical deployment. Korean fluency and prosody share a source target. SO denotes SpeechOcean ($n = 500$), Korean has $n = 636$, and J-AESOP has $n = 36$ development recordings; the small J-AESOP sample limits precision.

Table 4: Public-development Spearman correlations for the S condition.

| System | Accuracy | | | Fluency | | | Prosody | | |
|---|---|---|---|---|---|---|---|---|---|
| | SO | Korean | J-AESOP | SO | Korean | J-AESOP | SO | Korean | J-AESOP |
| speechocean | 0.816 | 0.699 | 0.967 | 0.886 | 0.668 | 0.856 | 0.890 | 0.653 | 0.929 |
| korean | 0.735 | 0.759 | 0.969 | 0.837 | 0.778 | 0.919 | 0.829 | 0.778 | 0.938 |
| aesop | 0.659 | 0.673 | 0.987 | 0.682 | 0.586 | 0.966 | 0.685 | 0.544 | 0.975 |
| koreanocean | 0.808 | 0.755 | 0.964 | 0.876 | 0.770 | 0.924 | 0.876 | 0.771 | 0.949 |
| aesopocean | 0.807 | 0.715 | 0.970 | 0.881 | 0.728 | 0.952 | 0.878 | 0.747 | 0.971 |
| koreanaesop | 0.683 | 0.755 | 0.976 | 0.834 | 0.779 | 0.959 | 0.830 | 0.777 | 0.973 |
| aesopkoreanocean | 0.804 | 0.748 | 0.971 | 0.875 | 0.769 | 0.953 | 0.872 | 0.770 | 0.974 |

# D Korean supervision and representation sensitivity

## D.1 Korean scoring and target mapping

Han, Chung, and Kim (2024; main-paper reference) define two 1–5 ratings: pronunciation accuracy and prosodic fluency. Accuracy addresses segmental clarity and errors; prosodic fluency combines stress, rhythm, intonation, speed, and pausing. The stored fields are `articulationScore` and `prosodyScore`. The second field supplies both fluency and prosody targets in the present implementation. Mapping $2.5(y-1)$ produces 0, 2.5, 5, 7.5, and 10 from the five integer levels; mapping does not equate the rubric with SpeechOcean or create an independent fluency annotation. Korean-only heads use the same features and targets for both constructs, yielding identical predictions in the evaluated configuration.

| Source | Mapped | Accuracy (paraphrased) | Prosodic fluency (paraphrased) |
|---|---|---|---|
| 1 | 0 | Extensive segmental difficulty; understanding is hard. | Strongly disrupted prosody, very slow delivery, excessive pauses. |
| 2 | 2.5 | Frequent errors substantially burden understanding. | Awkward prosody and slow delivery with prolonged, poorly placed pauses. |
| 3 | 5 | Recurring errors interfere with understanding. | Uneven prosody and rate, with frequent or misplaced breaks. |
| 4 | 7.5 | Minor errors with little intelligibility impact. | Mostly steady delivery with mild prosodic or pausing irregularities. |
| 5 | 10 | Clear segments without noticeable errors. | Natural prosody, rate, and phrase boundaries. |

## D.2 Ablation protocol

The protocol was recorded before fitting on October 18, 2026 (repository: `results/ablation-mac-v1/PROTOCOL.md`). The completed public-development sweep crossed seven training families, three constructs, and all 15 nonempty subsets of four feature blocks, giving 315 configurations and 945 corpus-level comparisons. X denotes XEUS, R XLS-R, P PhoneticXeus embeddings, and I the 17 phone diagnostics; I still requires PhoneticXeus inference, so omitting P does not necessarily avoid that encoder. Existing speaker-disjoint partitions, corpus-balanced weights, training-only standardization, and block gains were preserved. Each configuration selected a penalty from $\{0.001, 0.01, 0.1, 1, 10, 100\}$ using equal-weight mean Spearman across all three development corpora, breaking ties by lower macro MAE and then larger penalty. This differs from the earlier included-corpus tuning; no corpus in these tables is an untouched cross-corpus test. Private recordings and ratings were excluded from fitting and selection.

Each candidate was compared with the fixed-feature baseline refitted under the same sweep protocol (X+P+I for accuracy, X+R otherwise). Candidates within 0.01 of the numerical best were shortlisted. Promotion required macro Spearman gain at least 0.01, no corpus correlation loss exceeding 0.02, no corpus MAE increase exceeding 0.10 on the 0–10 scale, and a positive lower 95% paired speaker-bootstrap bound (2,000 draws, seed 42). Fewer blocks were preferred where benefits were inconclusive. These screening gates do not constitute correction for all search comparisons. Deployed heads were not automatically replaced.

The decision table distinguishes numerical maxima from candidates passing the complete gate. Two AESOP-only alternatives passed: X+P for fluency and X+P+I for prosody. Their macro gains over the refitted AESOP baselines were 0.0341 (95% interval [0.0118, 0.0619]) for fluency and 0.0389 ([0.0161, 0.0638]) for prosody. Both exceed the 0.01 within-family threshold. Neither became the overall construct selection, and neither was substituted into the external seven-family evaluation. The overall retained choices were koreanocean X+P+I accuracy and aesopkoreanocean X+R fluency/prosody. Thus retaining the overall configurations does not mean that no individual family improved. XEUS-only macro correlations for the selected families were 0.8407 for koreanocean accuracy and 0.8644/0.8680 for aesopkoreanocean fluency/prosody, compared with 0.8424 and 0.8660/0.8721 for the retained combinations. These small development differences do not establish that all additional blocks are necessary; no XEUS-only private evaluation or matched resource benchmark is reported. The retained choices reflect the recorded development rule, not independently established representation optimality. Matrices report all 315 macro correlations; the accompanying CSV preserves all 945 corpus-specific correlations and MAEs.

| Family | Construct | Numerical best | Best $\rho$ | Retained/gated | Promotion |
|---|---|---|---|---|---|
| speechocean | accuracy | X+P+I | 0.8276 | X+P+I | No |
| speechocean | fluency | X+P+I | 0.8655 | X+R | No |
| speechocean | prosody | X+I | 0.8640 | X+R | No |
| korean | accuracy | X+R+I | 0.8322 | X+P+I | No |
| korean | fluency | X+I | 0.8544 | X+R | No |
| korean | prosody | X | 0.8560 | X+R | No |
| aesop | accuracy | X+P+I | 0.7813 | X+P+I | No |
| aesop | fluency | X+P | 0.7961 | X+P | Yes |
| aesop | prosody | X+P+I | 0.7893 | X+P+I | Yes |
| koreanocean | accuracy | X+P+I | 0.8424 | X+P+I | No |
| koreanocean | fluency | X+P | 0.8685 | X+R | No |
| koreanocean | prosody | X+P+I | 0.8691 | X+R | No |
| aesopocean | accuracy | X+P+I | 0.8309 | X+P+I | No |
| aesopocean | fluency | X+R+P+I | 0.8654 | X+R | No |
| aesopocean | prosody | X+R+P+I | 0.8653 | X+R | No |
| koreanaesop | accuracy | X+R | 0.8337 | X+P+I | No |
| koreanaesop | fluency | X+R+P | 0.8600 | X+R | No |
| koreanaesop | prosody | X+R | 0.8599 | X+R | No |
| aesopkoreanocean | accuracy | X+R+P+I | 0.8425 | X+P+I | No |
| aesopkoreanocean | fluency | X+R+P+I | 0.8710 | X+R | No |
| aesopkoreanocean | prosody | X+R | 0.8721 | X+R | No |

### D.3 Representation matrices

Columns S, K, A, SK, SA, KA, and SKA denote SpeechOcean, Korean, J-AESOP, and their respective combinations. Values are macro development Spearman correlations; X, R, P, and I denote XEUS, XLS-R, PhoneticXeus embeddings, and IPA diagnostics.

Table 6: Representation ablation by construct and training family.

| Features | S | K | A | SK | SA | KA | SKA |
|---|---|---|---|---|---|---|---|
| **Accuracy** | | | | | | | |
| X | 0.8208 | 0.8321 | 0.7655 | 0.8407 | 0.8226 | 0.8318 | 0.8336 |
| R | 0.6715 | 0.7460 | 0.3186 | 0.7646 | 0.7465 | 0.7766 | 0.7972 |
| P | 0.7860 | 0.7366 | 0.7292 | 0.8228 | 0.8065 | 0.7600 | 0.8273 |
| I | 0.6080 | 0.6742 | 0.6201 | 0.6961 | 0.3225 | 0.5164 | 0.4372 |
| X+R | 0.7957 | 0.8297 | 0.7414 | 0.8322 | 0.8257 | 0.8337 | 0.8361 |
| X+P | 0.8190 | 0.8030 | 0.7732 | 0.8350 | 0.8231 | 0.8159 | 0.8354 |
| X+I | 0.8253 | 0.8321 | 0.7522 | 0.8411 | 0.8249 | 0.8149 | 0.8362 |
| R+P | 0.7505 | 0.7690 | 0.6650 | 0.8192 | 0.8008 | 0.7871 | 0.8287 |
| R+I | 0.7290 | 0.7875 | 0.5625 | 0.7968 | 0.7765 | 0.7830 | 0.8085 |
| P+I | 0.7991 | 0.7750 | 0.7582 | 0.8308 | 0.8155 | 0.7171 | 0.8356 |
| X+R+P | 0.7962 | 0.8104 | 0.7453 | 0.8280 | 0.8235 | 0.8225 | 0.8350 |
| X+R+I | 0.7984 | 0.8322 | 0.7223 | 0.8363 | 0.8266 | 0.8183 | 0.8382 |
| X+P+I | 0.8276 | 0.8212 | 0.7813 | 0.8424 | 0.8309 | 0.8048 | 0.8409 |
| R+P+I | 0.7723 | 0.7932 | 0.7065 | 0.8260 | 0.8104 | 0.7740 | 0.8381 |
| X+R+P+I | 0.8048 | 0.8218 | 0.7565 | 0.8363 | 0.8304 | 0.8132 | 0.8425 |
| **Fluency** | | | | | | | |
| X | 0.8551 | 0.8543 | 0.7782 | 0.8630 | 0.8588 | 0.8545 | 0.8644 |
| R | 0.7312 | 0.7734 | 0.4373 | 0.8213 | 0.8000 | 0.8100 | 0.8319 |
| P | 0.8428 | 0.7912 | 0.7640 | 0.8592 | 0.8495 | 0.8127 | 0.8599 |
| I | 0.6288 | 0.6757 | 0.6577 | 0.6876 | 0.4060 | 0.5606 | 0.5002 |
| X+R | 0.8364 | 0.8444 | 0.7620 | 0.8567 | 0.8599 | 0.8574 | 0.8660 |
| X+P | 0.8613 | 0.8465 | 0.7961 | 0.8685 | 0.8645 | 0.8578 | 0.8702 |
| X+I | 0.8569 | 0.8544 | 0.7671 | 0.8618 | 0.8576 | 0.8538 | 0.8662 |
| R+P | 0.8178 | 0.8162 | 0.7153 | 0.8555 | 0.8498 | 0.8415 | 0.8653 |
| R+I | 0.7833 | 0.8076 | 0.6342 | 0.8407 | 0.8291 | 0.8200 | 0.8458 |
| P+I | 0.8469 | 0.8099 | 0.7748 | 0.8570 | 0.8511 | 0.8004 | 0.8593 |
| X+R+P | 0.8481 | 0.8458 | 0.7730 | 0.8609 | 0.8654 | 0.8600 | 0.8696 |
| X+R+I | 0.8443 | 0.8516 | 0.7400 | 0.8593 | 0.8598 | 0.8533 | 0.8652 |
| X+P+I | 0.8655 | 0.8508 | 0.7866 | 0.8666 | 0.8644 | 0.8498 | 0.8677 |
| R+P+I | 0.8340 | 0.8283 | 0.7268 | 0.8576 | 0.8515 | 0.8333 | 0.8653 |
| X+R+P+I | 0.8509 | 0.8481 | 0.7635 | 0.8639 | 0.8654 | 0.8556 | 0.8710 |
| **Prosody** | | | | | | | |
| X | 0.8631 | 0.8560 | 0.7817 | 0.8688 | 0.8599 | 0.8573 | 0.8680 |
| R | 0.7489 | 0.7740 | 0.3583 | 0.8172 | 0.7891 | 0.8178 | 0.8377 |
| P | 0.8313 | 0.7768 | 0.7520 | 0.8503 | 0.8416 | 0.7970 | 0.8526 |
| I | 0.6228 | 0.6834 | 0.6322 | 0.6960 | 0.4179 | 0.5379 | 0.5035 |
| X+R | 0.8410 | 0.8483 | 0.7504 | 0.8657 | 0.8653 | 0.8599 | 0.8721 |
| X+P | 0.8620 | 0.8384 | 0.7872 | 0.8664 | 0.8615 | 0.8508 | 0.8681 |
| X+I | 0.8640 | 0.8524 | 0.7627 | 0.8659 | 0.8609 | 0.8499 | 0.8693 |
| R+P | 0.8077 | 0.8070 | 0.6886 | 0.8517 | 0.8409 | 0.8296 | 0.8586 |
| R+I | 0.7789 | 0.8076 | 0.5893 | 0.8412 | 0.8229 | 0.8208 | 0.8498 |
| P+I | 0.8361 | 0.7979 | 0.7792 | 0.8527 | 0.8449 | 0.7793 | 0.8539 |
| X+R+P | 0.8444 | 0.8408 | 0.7602 | 0.8662 | 0.8647 | 0.8540 | 0.8707 |
| X+R+I | 0.8485 | 0.8512 | 0.7281 | 0.8663 | 0.8639 | 0.8495 | 0.8687 |
| X+P+I | 0.8635 | 0.8459 | 0.7893 | 0.8691 | 0.8649 | 0.8420 | 0.8690 |
| R+P+I | 0.8235 | 0.8210 | 0.7086 | 0.8522 | 0.8437 | 0.8214 | 0.8608 |
| X+R+P+I | 0.8485 | 0.8455 | 0.7554 | 0.8668 | 0.8653 | 0.8477 | 0.8715 |

## E Execution environment

The same Windows/WSL machine used for the reported capacity test was inspected on October 19, 2026. The installation reports Ubuntu 26.04.1 LTS, kernel `6.18.54.1-microsoft-standard-WSL2`, Python 3.11.17, and

PyTorch `2.11.0+cu128` (CUDA build 12.8). Hardware comprises an Intel Core i7-11700K with 16 logical CPUs and an NVIDIA RTX 3080 with 10 GiB VRAM; the host has 32 GB RAM, with approximately 24 GiB allocated to WSL and 8 GiB swap. The reported inference capacity test used the CPU, not CUDA. These versions document the inspected installation; the archived timing and memory trace, rather than a new benchmark, supplies the reported performance measurements.

# F External rater backgrounds and procedure

The original human-evaluation study reports three experienced university raters. The Korea-based rater had more than 20 years of teaching experience, including a decade of university pronunciation instruction and speaking assessment. The Japan-based rater had nearly 30 years of teaching experience and more than 10 years as a Cambridge English speaking assessor. The China-based rater had nearly 20 years of experience, including high-stakes speech rating and university speaking assessment in an international joint-degree program. All three held professorial positions, as confirmed by the study author.

Before evaluating the source recordings, the raters jointly assessed separate Stella recordings spanning proficiency levels and discussed discrepancies to establish shared criteria for segmental accuracy, fluency, and prosody. They then scored the 306 source recordings independently in randomized order, blinded to participant identity and automated scores. Ratings used a 0–100 scale in five-point increments, with communicative-effectiveness anchors labelled "High Intelligibility" above 85 and "Low Intelligibility" below 40. These are the documented general scale anchors, not a newly constructed analytic rubric for each dimension. The original study's authors served as its raters; their ratings were reused as the external reference for the present model evaluation. One recording was subsequently excluded for excessive duration, leaving the matched 305-recording cohort. Human calibration used separate recordings and did not fit or calibrate any automated scorer.